

vint Cerf

1964

TCP / IP

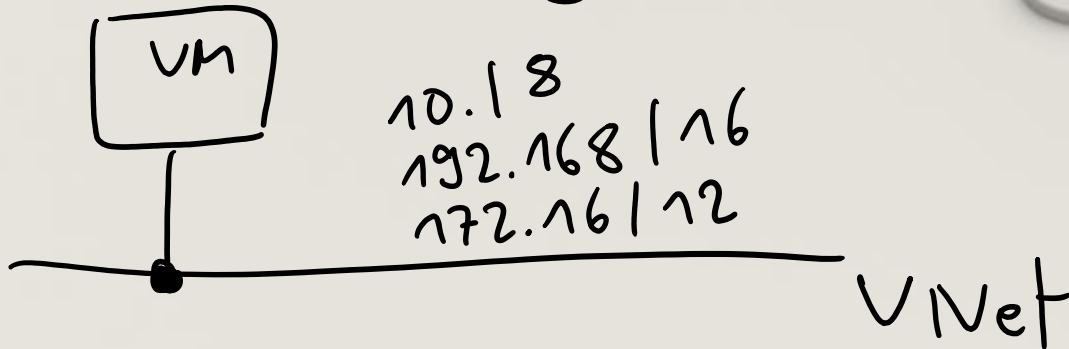
32 Bit

IPv6 — pub
— priv

SDN

AZ-104

Administer Virtual Networking



AZ-104 Agenda

- 01: Administer Identity
- 02: Administer Governance and Compliance
- 03: Administer Azure Resources
- 04: Administer Virtual Networking ←
- 05: Administer Intersite Connectivity
- 06: Administer Network Traffic Management
- 07: Administer Azure Storage
- 08: Administer Azure Virtual Machines
- 09: Administer PaaS Compute Options
- 10: Administer Data Protection
- 11: Administer Monitoring

Learning Objectives - Administer Virtual Networking

- Configure Virtual Networks
- Configure Network Security Groups
- Host your domain on Azure DNS
- Lab 04 – Implement Virtual Networks

Wins

Configure Virtual Networks



Learning Objective - Configure Virtual Networks

- Review Virtual Network Fundamentals
- Review IP Addressing Fundamentals
- Plan for IP Addressing
- Review Examples of Public IP Addressing
- Demonstration – Virtual Networks and IP addressing
- Learning Recap

Exam objectives

Configure and manage virtual networks in Azure

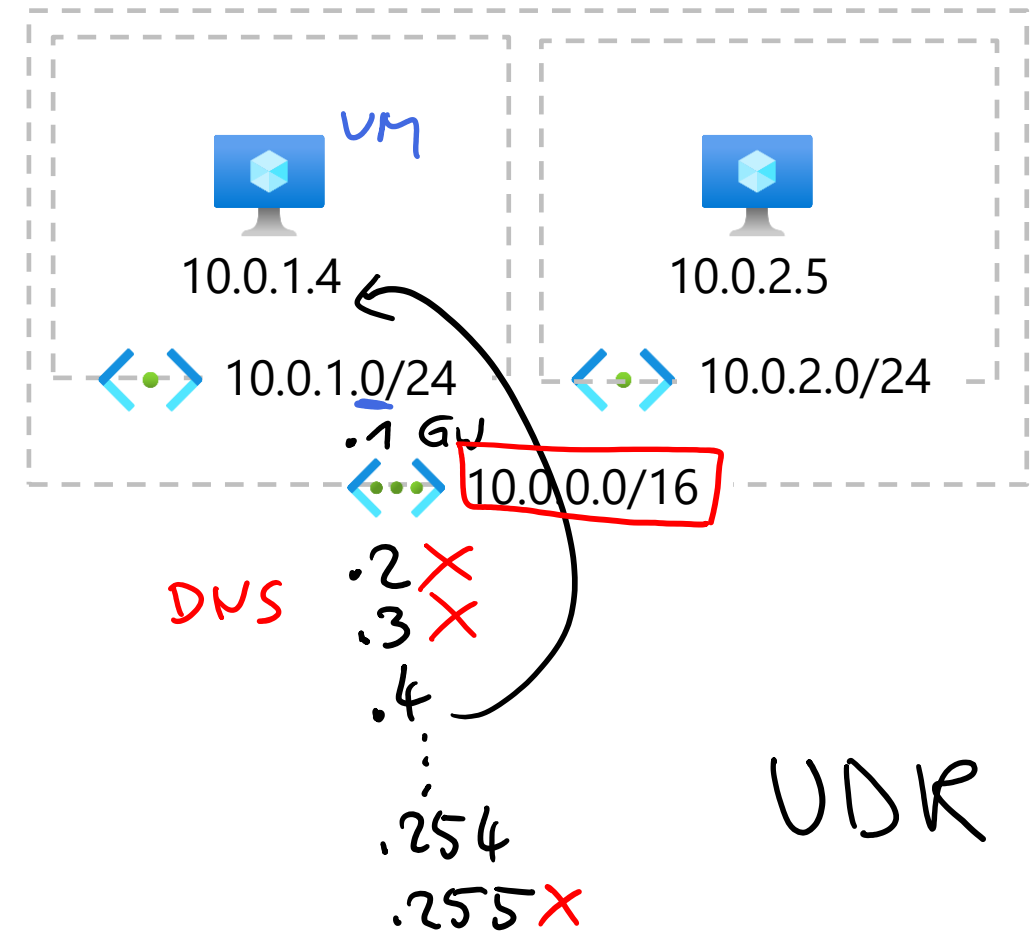
- Create and configure virtual networks and subnets
- Configure public IP addresses

Review Virtual Network Fundamentals

- **Virtual Network (VNet)**: The main boundary, a large private IP address space (e.g., 10.0.0.0/16) representing your isolated cloud network.
- **Subnets**: Smaller divisions within the VNet (e.g., 10.0.1.0/24, 10.0.2.0/24) for grouping resources, often public (with gateways) or private (for databases).
- **IP Addresses**: Individual addresses assigned to resources (VMs, databases) within subnets, following the subnet's range (e.g., 10.0.1.4, 10.0.2.5).
- **CIDR Notation**: /16, /24 notation showing network size and available IP addresses.

/28

<...> 10.1.0.0/16
└ Sub 1
 Sub 2
/ Peering

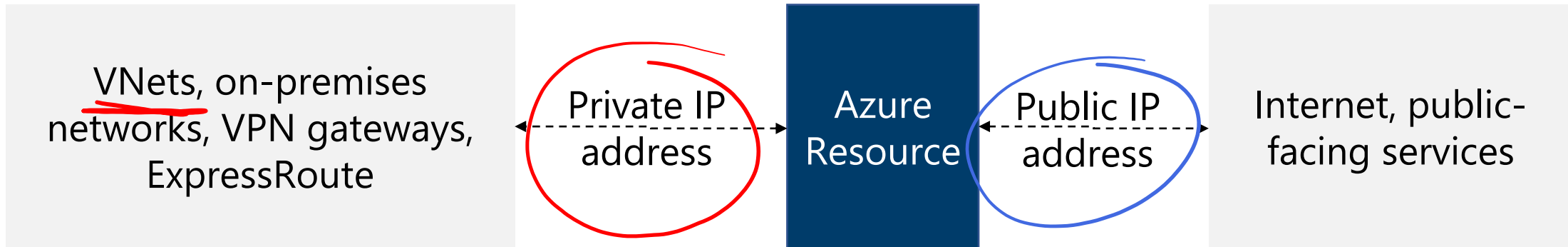


Review IP Addressing Fundamentals

- Not all IP addresses are available for assignment.
- The first four and the last IP address in every subnet is reserved for internal platform use (network/broadcast, default gateway, Azure DNS).
- Multicast, Broadcast, Loopback, Link-Local, and Internal DNS IP address ranges are not available for assignment.
- RFC 1918 defines specific IPv4 address ranges for private networks. These addresses are not routed on the public internet.

10.0.0.0/16		
10.0.0.0 /16		
10.0.0.0 - 10.0.255.255 65,536 addresses		
Subnets	IP address range	Size
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)

Plan for Azure IP Addressing

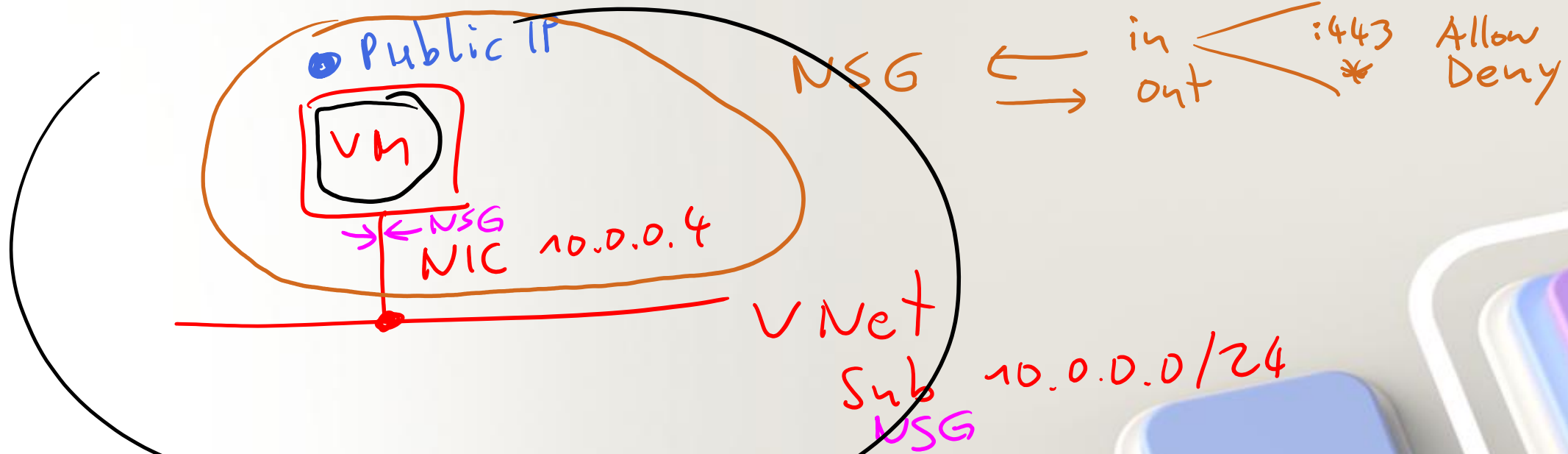


- Ensure address spaces don't overlap
- Use a few large virtual networks instead of multiple smaller ones
- Decide if your IP addresses should be private or public
- Decide if the IP addresses should be statically or dynamically allocated – consider using separate subnets
- Subnets shouldn't cover the entire address space of the virtual network
- Never manage IP addresses from within the virtual machine

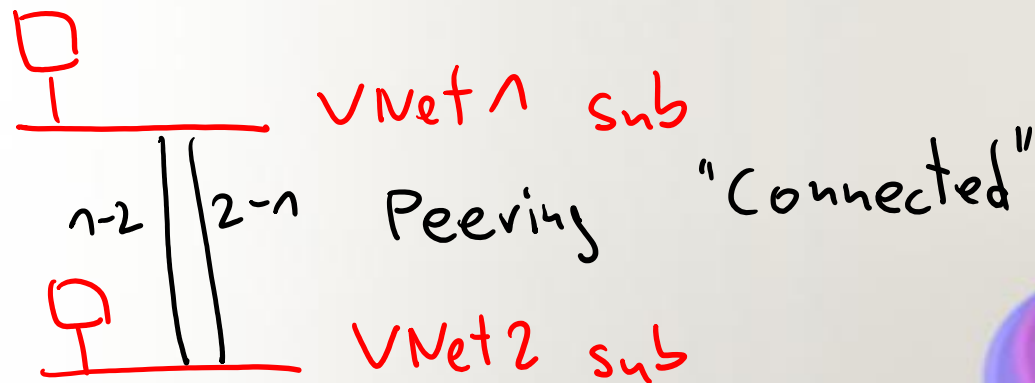
Review Examples of Public IP Addressing

- Public IP addresses allow Internet resources to communicate inbound to Azure resources
- Enables communication to the internet and public-facing Azure services
- Dedicated exclusively to the resource until you unassign it or delete the resource
- Public IP address prefixes (a reserved range of public IP addresses in Azure) are also available.

Public IP addresses	IP address association
Virtual Machine 	NIC
Load Balancer	Front-end configuration
VPN Gateway	Gateway IP configuration
Application Gateway	Front-end configuration
Azure Firewall	Front-end configuration



Configure Network Security Groups (NSGs)



Learning Objectives – Network Security Groups

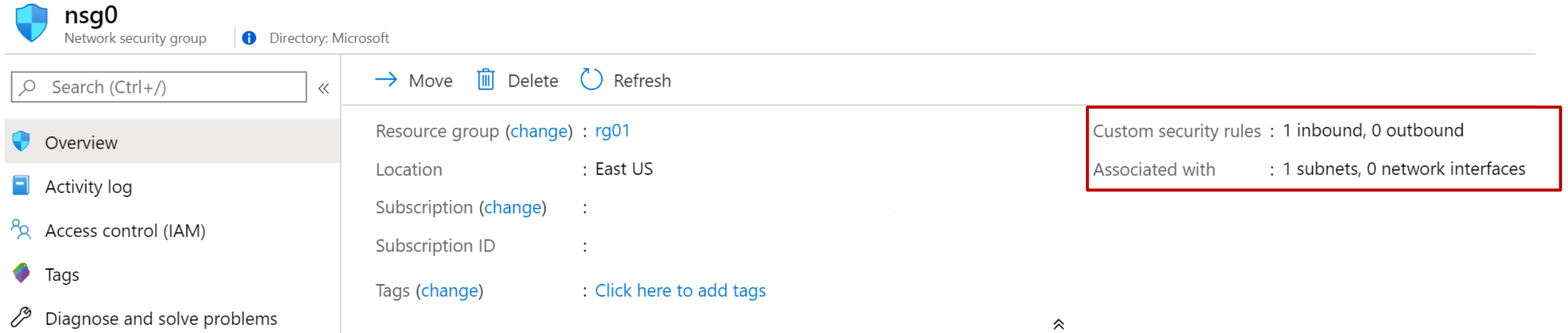
- Implement Network Security Groups
- Determine NSG Rules
- Determine NSG Effective Rules
- Create NSG Rules
- Implement Application Security Groups
- Demonstration – NSGs
- Learning Recap

Exam objectives

Configure secure access to virtual networks

- Create and configure network security groups and application security groups
- Evaluate effective security rules in NSGs

Implement Network Security Groups



nsg0
Network security group | Directory: Microsoft

Search (Ctrl+/) << → Move Delete Refresh

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems

Resource group ([change](#)) : rg01
Location : East US
Subscription ([change](#)) :
Subscription ID :
Tags ([change](#)) : [Click here to add tags](#)

Custom security rules : 1 inbound, 0 outbound
Associated with : 1 subnets, 0 network interfaces

- Limits network traffic to resources in a virtual network
- Lists the security rules that allow or deny inbound or outbound network traffic
- Associated to a subnet or a network interface
- Can be associated multiple times

Determine NSG Rules

Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✔ Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	✔ Allow
65500	DenyAllInBound	Any	Any	Any	Any	✘ Deny

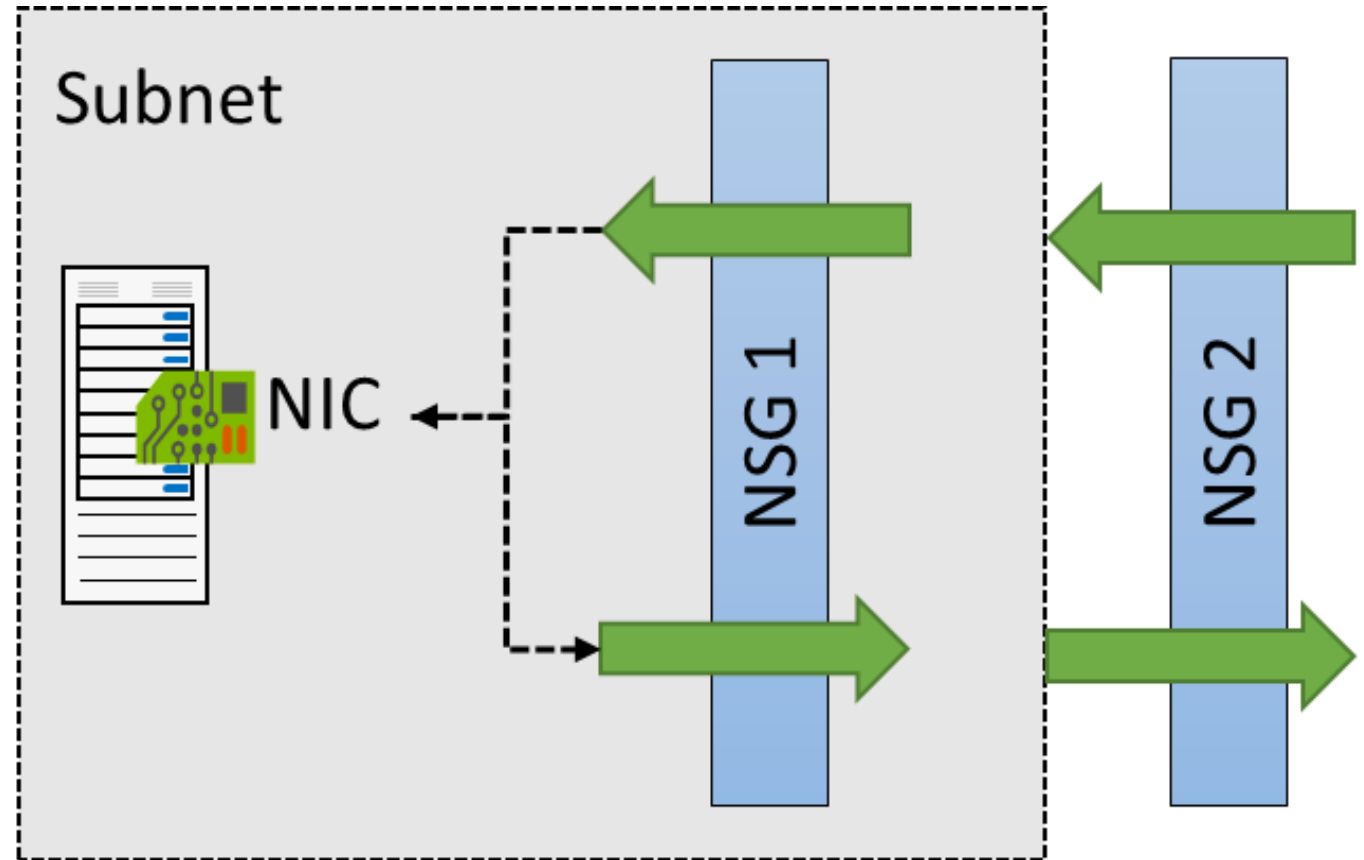
Outbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	✔ Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	✔ Allow
65500	DenyAllOutBound	Any	Any	Any	Any	✘ Deny

- Security rules in NSGs enable you to filter network traffic that can flow in and out of virtual network subnets and network interfaces
- There are default security rules. You cannot delete the default rules, but you can add other rules with a higher priority

Determine NSG Effective Rules

- NSGs are evaluated independently for the subnet and NIC
- An "allow" rule must exist at both levels for traffic to be admitted
- Use the Effective Rules link if you are not sure which security rules are being applied



Network Interface: **vm01990**

[Effective security rules](#)

[Topology](#)

Virtual network/subnet: **vnet01/subnet0**

NIC Public IP: -

NIC Private IP: **10.1.0.4**

Accelerated networking: **Disabled**

Create NSG rules

- For each rule, you can specify source and destination details, port, and protocol.
- Select from many preconfigured services, or create your own custom rules
- Rules are processed in priority order. The lower the number, the higher the priority.
- There are limits on the number of rules per NSG for each Azure location and subscription.

Add inbound security rule

Source ⓘ
Any

Source port ranges * ⓘ
*

Destination ⓘ
Any

Service ⓘ
Custom

Destination port ranges * ⓘ
8080

Protocol
☒ Any ☐ UDP
☐ TCP ☐ ICMPv4

Action
☒ Allow ☐ Deny

Priority * ⓘ
100

Name *
AllowAnyCustom8080Inbound

Service ⓘ

Custom

HTTP

HTTPS

SSH

RDP

MS SQL

MySQL

PostgreSQL

Windows Admin Center

Custom

FTP

SMTP

DNS (TCP)

DNS (UDP)

POP3

IMAP

LDAP

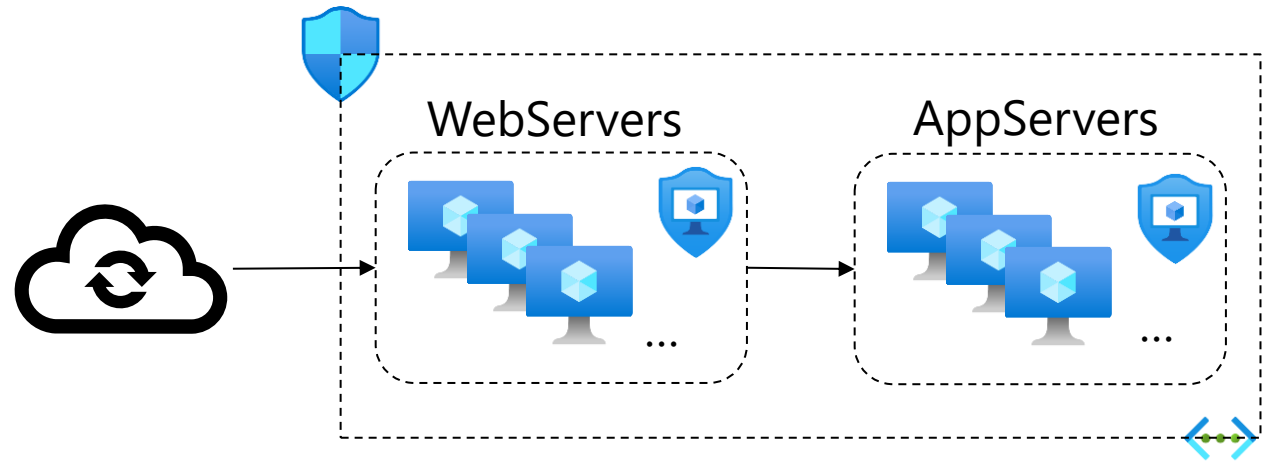
SMTPS

IMAPS

POP3S

Implement Application Security Groups (ASG)

- Group VMs by application or workload and apply network security policies to those groups - instead of to individual IP addresses.
- ASGs must be in the same region and subscription as the VMs they group.
- ASGs can be the source or destination in a security rule



Action	Source	Destination	Port
Allow	Internet	WebServers	80,443
Allow	WebServers	AppServers	443
Deny	Any	Any	Any

Host your domain on Azure DNS



Learning Objective - Host your domain on Azure DNS

- Review DNS Fundamentals
- What is Azure DNS?
- Create a Public DNS Zone and a DNS Record Set
- Configure a Private DNS zone
- Demonstration – Azure DNS
- Learning Recap

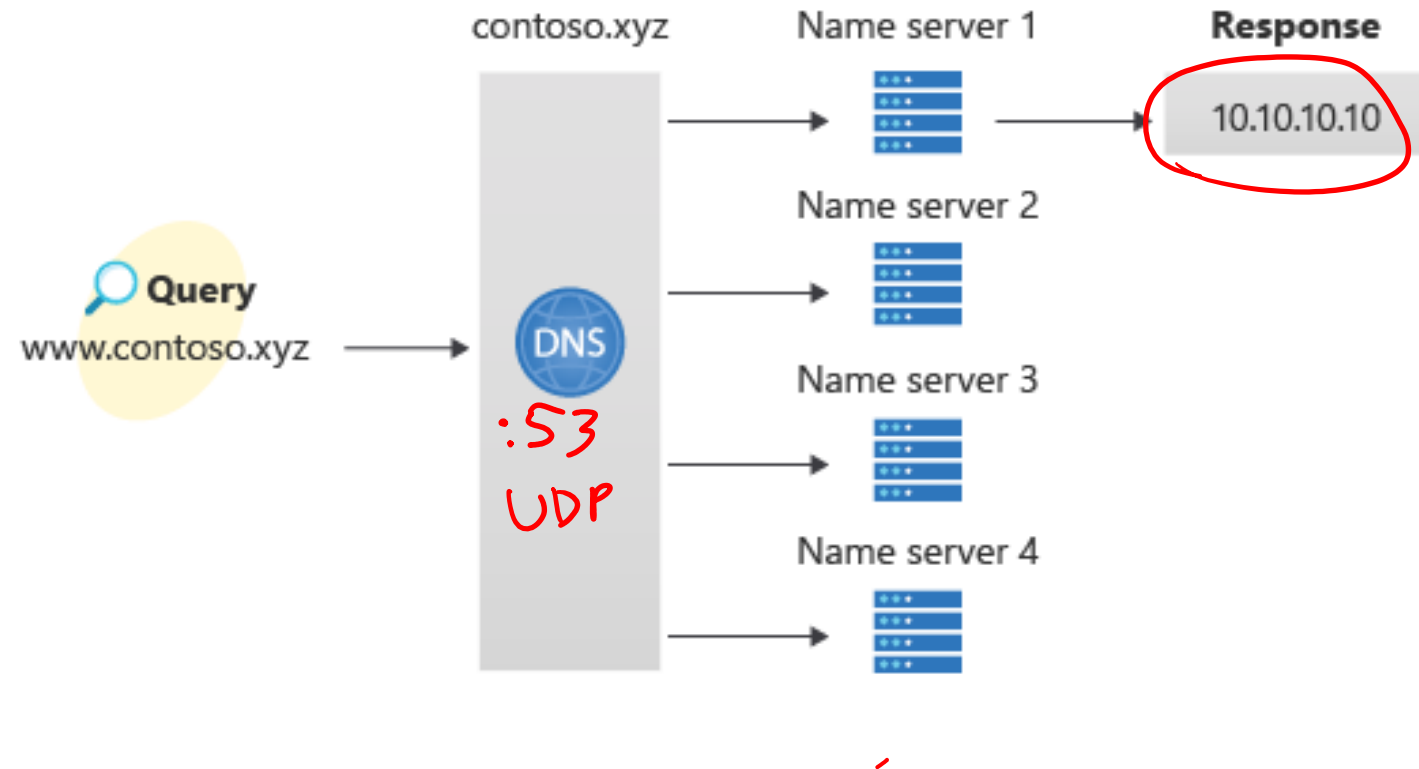
Exam objectives

Configure name resolution and load balancing

- Configure Azure DNS

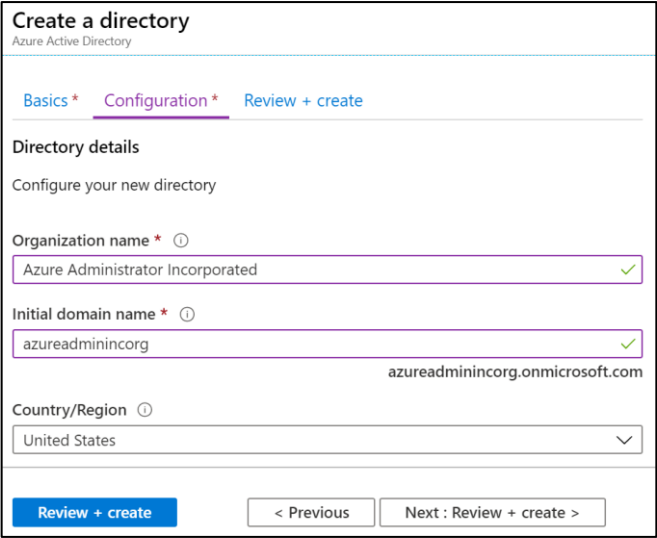
Review Domain Name Service (DNS) Fundamentals

- **DNS Zone:** Hosts the DNS records for a particular domain.
- **DNS Records:** Information to handle requests.
 - **A:** Maps a domain name to IPv4 address
 - **AAAA:** Maps a domain name to IPv6 address
 - **CNAME :** Maps for one domain to another (alias record)
- **DNS Resolution:** The process of translating a human-readable domain name into a machine-readable IP address.



What is Azure DNS?

- When you create a new tenant, a new default domain is created
- The domain has initial domain name in the form domainname.onmicrosoft.com △ EntveID
- You can add a custom domain name
- After the custom name is added it must be verified – this demonstrates ownership of the domain



Create a directory
Azure Active Directory

Basics * Configuration * Review + create

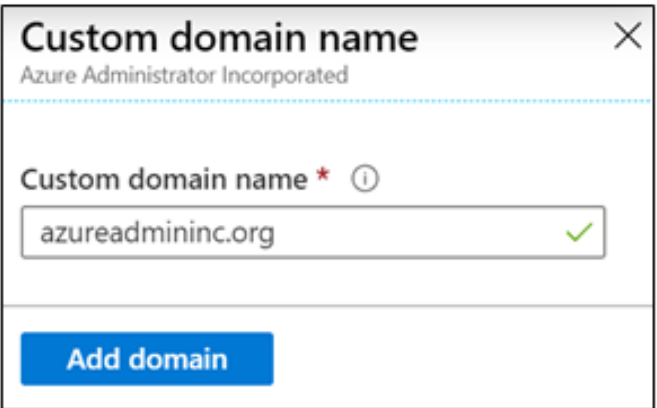
Directory details
Configure your new directory

Organization name * ⓘ
Azure Administrator Incorporated ✓

Initial domain name * ⓘ
azureadminincorg ✓
azureadminincorg.onmicrosoft.com

Country/Region ⓘ
United States ▼

Review + create < Previous Next : Review + create >



Custom domain name ✕
Azure Administrator Incorporated

Custom domain name * ⓘ
azureadmininc.org ✓

Add domain

Create a Public DNS Zone and a DNS Record Set

You must claim and register (GoDaddy, Namecheap, or Google Domains) your domain

IANA com
Contoso.com
GoDaddy

Create DNS zone

BasicsTagsReview + create

A DNS zone is used to host the DNS records for a particular domain. For example, the domain 'contoso.com' may contain a number of DNS records such as 'mail.contoso.com' (for a mail server) and 'www.contoso.com' (for a web site). Azure DNS allows you to host your DNS zone and manage your DNS records, and provides name servers that will respond to DNS queries from end users with the DNS records that you create. [Learn more](#).

Project details

Subscription *MSDN Platforms Subscription

Resource group *rg-dns
[Create new](#)

Instance details

Name *azureadmininc.org

Resource group location ⓘEast US

Review + create

Previous

Next : Tags >

[Download a template for automation](#)



Add record set

contosoces.com

Name

Type

A – IPv4 Address records

AAAA – IPv6 Address records

CAA – Certificate Authorities to authorize certificates

CNAME – Link your subdomain to another record

MX – Mail eXchange records

NS – Name Server records

SRV – Service records

TXT – Text records

PTR – Pointer records

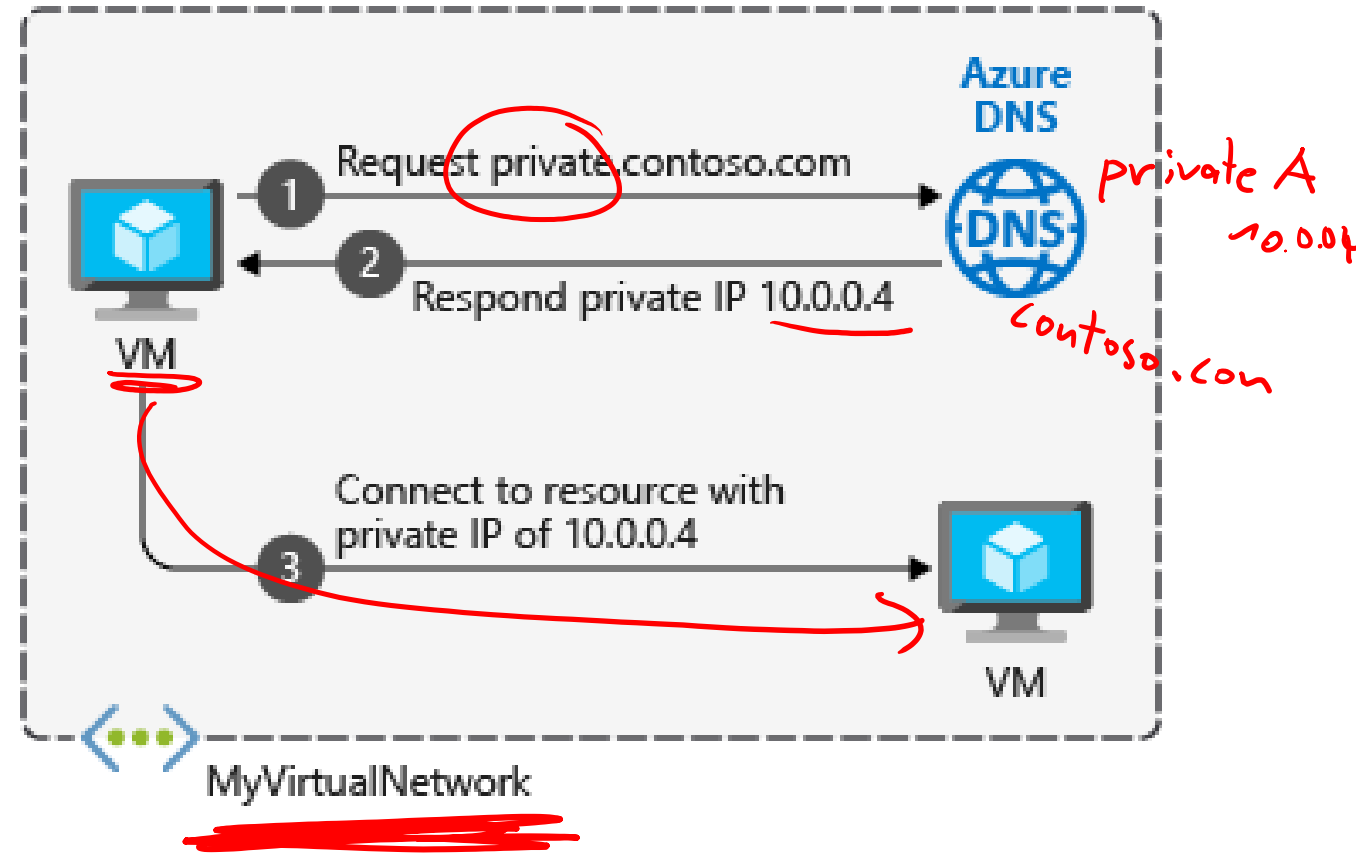
DS - Delegation Signer records

TLSA – TLS Authentication records

Configure a Private DNS Zone

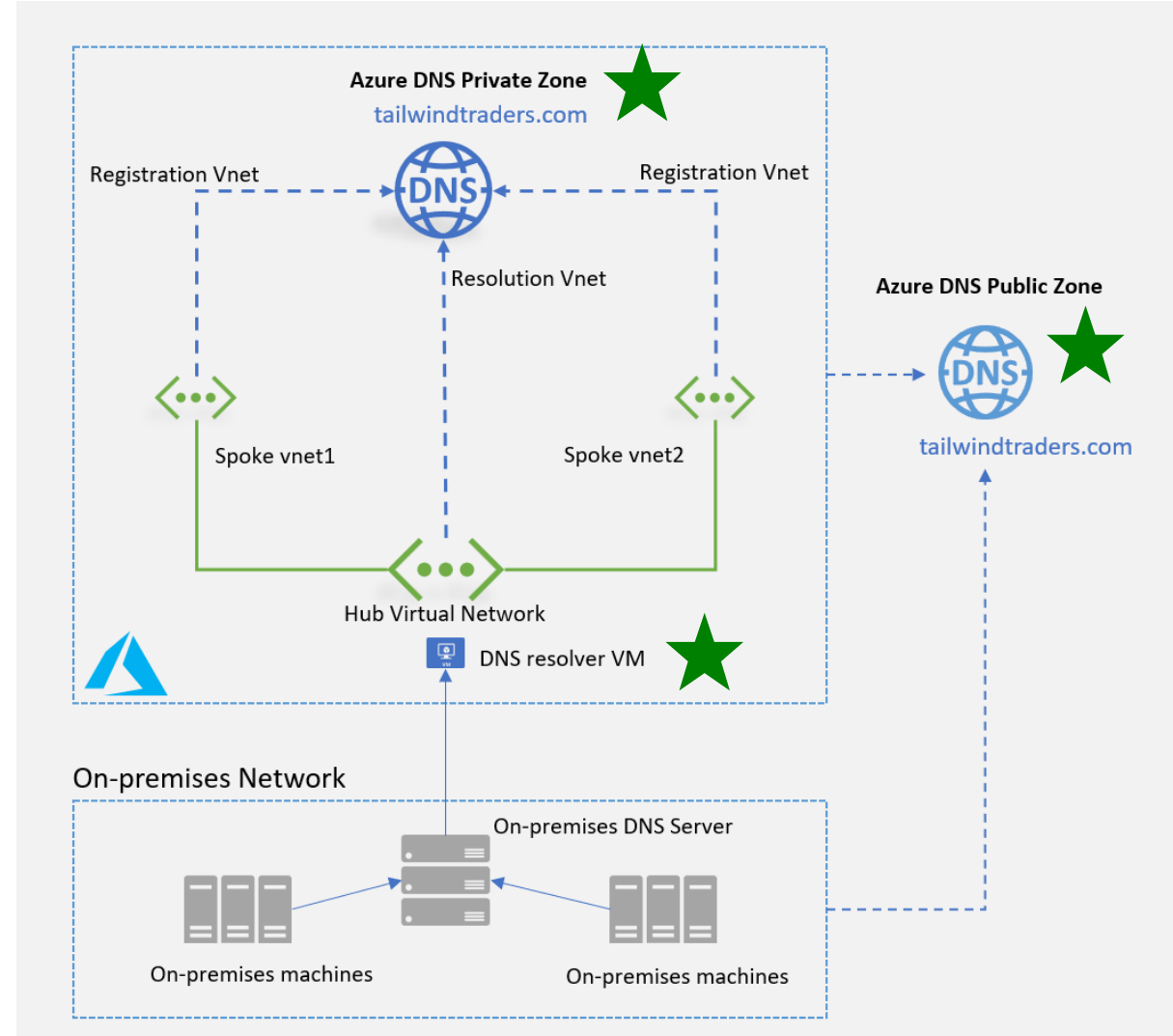
- Use your own domain names
- Access from any region, any subscription, any VNet, and any tenant
- Provides name resolution for VMs within a VNet and between VNets
- Use auto registration to populate the DNS records for VMs in the zone
- Be sure to create a link from the private zone to the virtual network

Example: Name resolution scoped to a single virtual network



Understand Azure DNS Capabilities

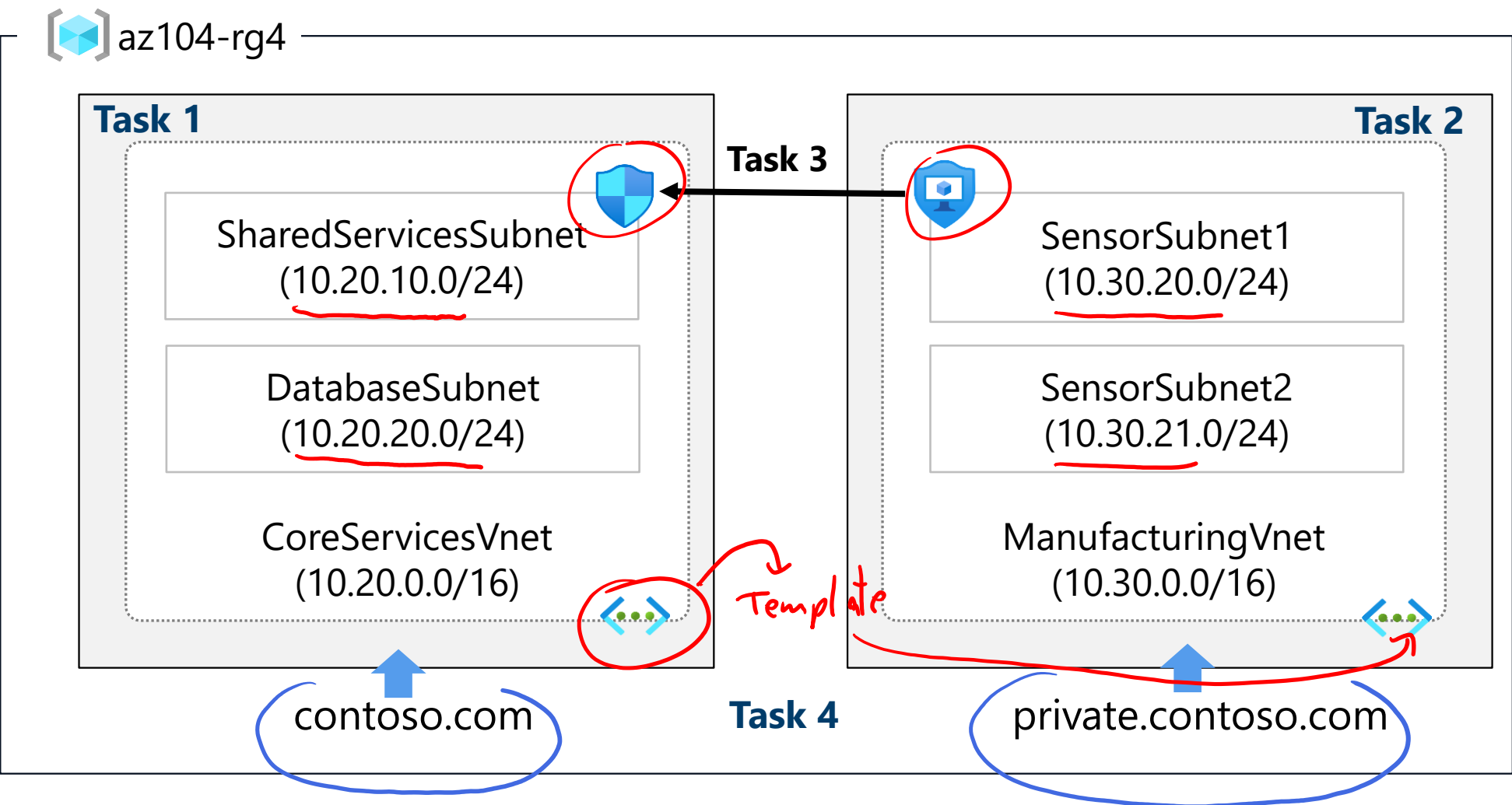
- **Azure Public DNS** is a hosting service that provides name resolution by using Microsoft Azure infrastructure.
- **Azure Private DNS** manages and resolves domain names in the virtual network without the need to configure a custom DNS solution.
- **Azure DNS Private Resolver** resolves DNS queries for private DNS zones from anywhere.



Lab – Implement Virtual Networks



Lab 04 – Architecture diagram



End of presentation



